
LLSMF2904 - Managing Complexity

Final Assignment

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1 Division of Tasks

The first step in our project was to assign roles and responsibilities to ensure efficiency and accountability. We appointed Jean as the team leader, entrusting him with the responsibility of overseeing the project's progress, ensuring the timely submission of our final report, and serving as the main point of contact with the teacher in case of any issues or clarifications.

Johanna worked on the first part of the project, ensuring that the foundational aspects were well developed. Throughout the process, she kept the rest of the team informed about the progress, allowing for constructive feedback and necessary adjustments. Evrard was responsible for solving exercise 1 of part 2 focusing on accuracy and clarity. Meanwhile, Jean took charge of exercises 2 of part 2 and Victor of the 3rd part of part 2, ensuring a thorough and well-structured approach to problem-solving. Once all individual contributions were completed, everyone helped compiling and organizing the entire team's work into a cohesive and professionally formatted final report.

Given the collaborative nature of the project, effective communication was crucial. To facilitate this, we established a system of weekly meetings where each team member provided updates on their progress. These meetings served as an opportunity to discuss any challenges, offer feedback, and refine our work to improve overall quality. This structured approach not only helped us stay on track but also ensured that our final submission met high standards of clarity, coherence, and professionalism.

2 Part 1

2.1 Weekly Returns, Means and Standard Deviation

To evaluate the performance and risk associated with the different assets, we computed the weekly returns for each of the five series: Stock Market SP500 (SM), Nasdaq (N), Gold (G), Bitcoin (B), and Silver (S).

The **weekly return** was calculated using the following formula:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where:

- P_t represents the price at time t , and
- P_{t-1} represents the price at time $t - 1$ (the previous week).

Once the weekly returns were computed, we calculated the **mean return** using the geometric mean formula:

$$\bar{R}_g = \left(\prod_{t=1}^N (1 + R_t) \right)^{\frac{1}{N}} - 1$$

where N is the number of weekly returns. This measure provides a first indication of the average performance of each asset over the analyzed period.

The **standard deviation**, which measures the volatility of returns, was computed as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (R_i - \bar{R}_g)^2}$$

A higher standard deviation indicates higher risk and greater fluctuations in the asset's price.

Using the above developed formulas, we obtained the results summarized in Table 1.

Asset	Mean Return	Standard Deviation
Stock Market SP500 (SM)	0.00149	0.02276
Nasdaq (N)	0.00190	0.02746
Gold (G)	0.00157	0.02622
Bitcoin (B)	0.01697	0.14806
Silver (S)	0.00117	0.04278

Table 1: Mean and Standard Deviation of Weekly Returns

Interpretation

The results indicate that Bitcoin exhibits the highest average return (1.70%), significantly exceeding the returns of the other assets. Among the traditional assets, Nasdaq shows the highest average return (0.19%), followed by Gold (0.16%) and the SP500 (0.15%), while Silver records the lowest average return (0.12%).

In terms of risk, measured by the standard deviation of returns, Bitcoin is by far the most volatile asset (14.81%), indicating substantially higher fluctuations compared to the other markets. Among the remaining assets, Silver shows the second highest volatility (4.28%), followed by Nasdaq (2.75%), Gold (2.62%), and the SP500 (2.28%), which appears to be the most stable asset.

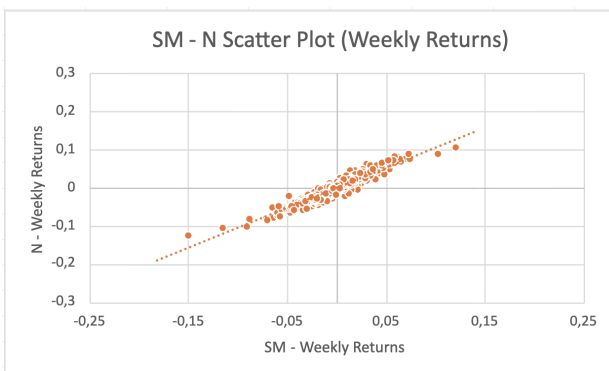
Overall, the results suggest that while Bitcoin offers the highest potential return, it also involves considerably higher risk. In contrast, the SP500 provides lower returns but also lower volatility, making it a relatively more stable investment compared to the other assets in the sample.

2.2 Scatter Plots of Weekly Returns

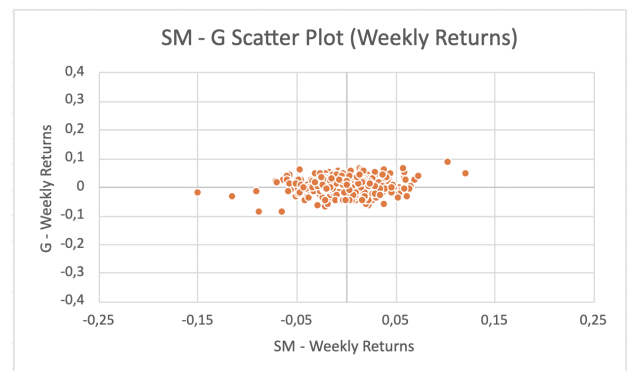
To examine the relationships between the different asset classes, we created scatter plots comparing the weekly returns of the following asset pairs:

- Stock Market SP500 (SM) vs. Nasdaq (N)
- Stock Market SP500 (SM) vs. Gold (G)
- Nasdaq (N) vs. Bitcoin (B)
- Gold (G) vs. Silver (S)

For each pair of assets, only overlapping time periods were used to ensure comparability of observations. These scatter plots provide insights into the dependence structures between these financial assets and allow for a visual assessment of potential correlations between their returns. For consistency and comparability with the monthly return plots (shown in 2.5), the same axis scales are used, which results in a slightly more zoomed-out appearance of the weekly scatter plots.

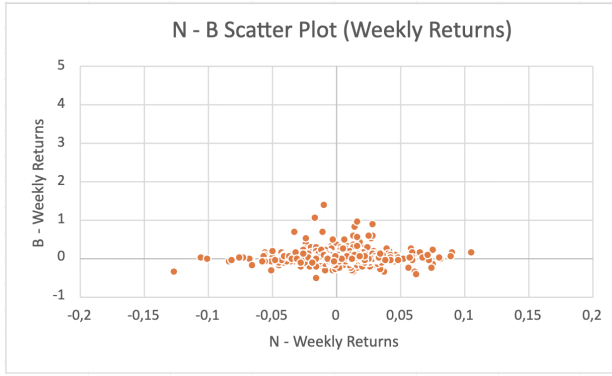


(a) SP500 vs Nasdaq

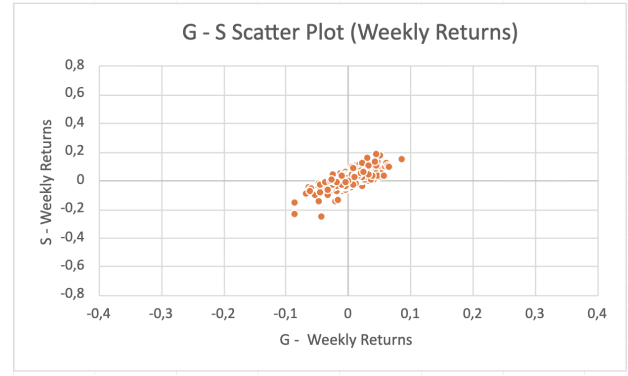


(b) SP500 vs Gold

Figure 1: Scatter plots of weekly returns for the different asset pairs (1/2)



(c) Nasdaq vs Bitcoin



(d) Gold vs Silver

Figure 1: Scatter plots of weekly returns for the different asset pairs (2/2)

Interpretation of Scatter Plots

The first scatter plot (Figure 1a) shows a strong positive correlation between the SP500 and the Nasdaq. The points are closely clustered along an upward sloping line, indicating that movements in the broader stock market tend to be closely mirrored by movements in the technology-focused Nasdaq index.

The second scatter plot (Figure 1b) displays a much weaker relationship between the SP500 and gold. The points appear more dispersed without a clear linear pattern, suggesting that gold price movements are not strongly linked to fluctuations in the stock market.

The third scatter plot (Figure 1c) illustrates the relationship between the Nasdaq and Bitcoin. The observations are widely scattered, indicating a relatively weak and unstable relationship between technology stocks and cryptocurrency returns.

Finally, the fourth scatter plot (Figure 1d) shows a clear positive relationship between gold and silver. The points follow an upward pattern, reflecting the fact that both assets belong to the precious metals market and tend to respond to similar economic factors.

Overall, the scatter plots indicate that stock indices such as the SP500 and Nasdaq move closely together, while assets like gold and Bitcoin behave more independently. Precious metals such as gold and silver exhibit strong co-movements, whereas assets from different classes (e.g., equities and cryptocurrencies) show weaker relationships. These patterns highlight the potential benefits of diversification when combining assets with low correlations in an investment portfolio.

2.3 Empirical Copula Plots of Weekly Returns

To observe the dependence structure between the return series and go beyond the simple scatter plots of the returns, we computed the empirical copulas for each pair of assets: Stock Market vs. Nasdaq, Stock Market vs. Gold, Nasdaq vs. Bitcoin, and Gold vs. Silver. The empirical copula captures the joint rank-based behavior of two variables.

Specifically, for two return series X and Y , we computed their pseudo-observations as:

$$U_i = \frac{\text{rank}(-X_i)}{n+1}, \quad V_i = \frac{\text{rank}(-Y_i)}{n+1}$$

This transformation maps each return into the unit interval $[0,1]$, preserving ordinal information while removing the influence of marginal distributions. As a result, the dependence structure between the assets can be analyzed independently of their individual distributions. As in the scatter plots, for each pair of assets, only overlapping time periods were used to ensure comparability of observations.

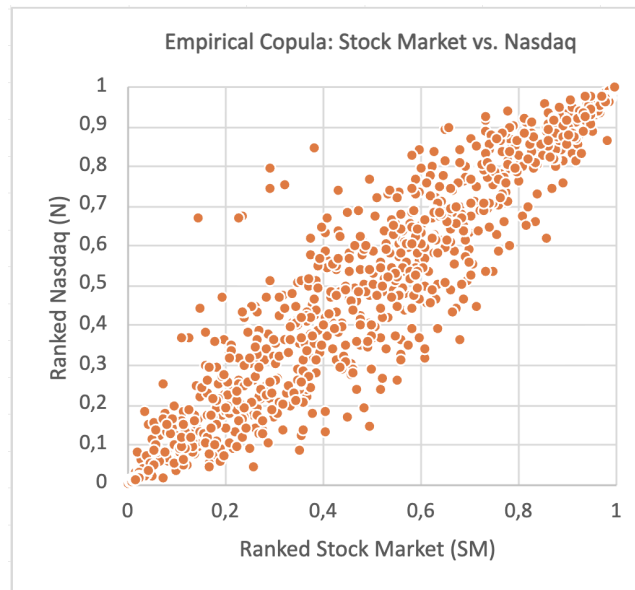


Figure 2: Empirical Copula: Stock Market vs Nasdaq

The copula between the Stock Market and Nasdaq clearly shows a strong positive dependence structure. Most observations lie close to the diagonal of the unit square, indicating that high (or low) returns in the stock market tend to occur together with similarly extreme returns in the Nasdaq. This strong co-movement reflects the close relationship between the two equity indices, which are both influenced by similar macroeconomic and financial market conditions.

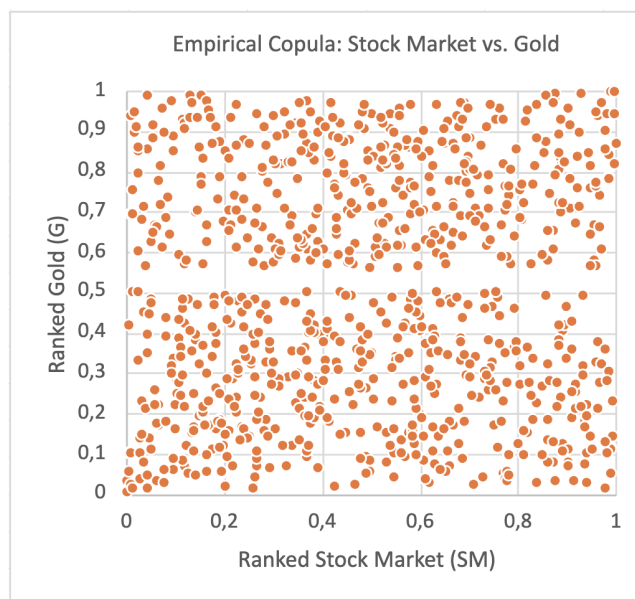


Figure 3: Empirical Copula: Stock Market vs Gold

In contrast, the copula between the Stock Market and Gold displays a much more diffuse distribution of points across the unit square. There is no clear diagonal pattern, suggesting

that the dependence between these two assets is relatively weak. This indicates that movements in gold prices are largely independent from stock market returns, which is consistent with gold often being considered a diversification asset in financial markets.

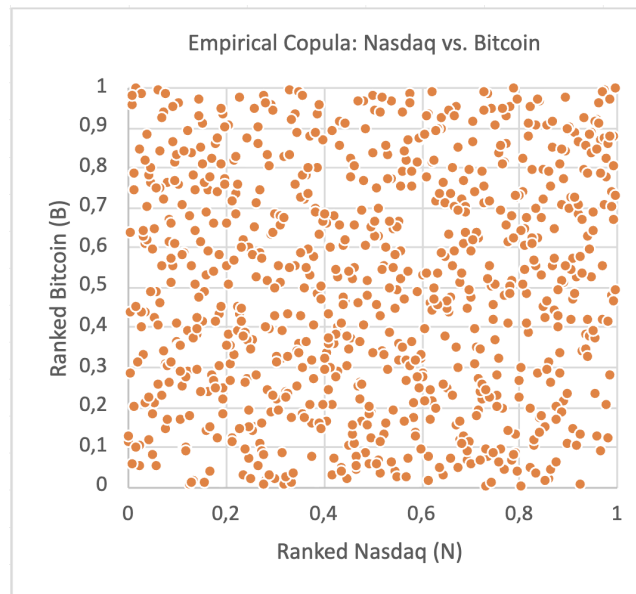


Figure 4: Empirical Copula: Nasdaq vs Bitcoin

The empirical copula between Nasdaq and Bitcoin also shows a rather scattered cloud of observations. The points are widely distributed throughout the unit square, without a clear concentration. This pattern suggests that the dependence between Nasdaq returns and Bitcoin returns is weak, indicating that the cryptocurrency market behaves largely independently from traditional equity markets.

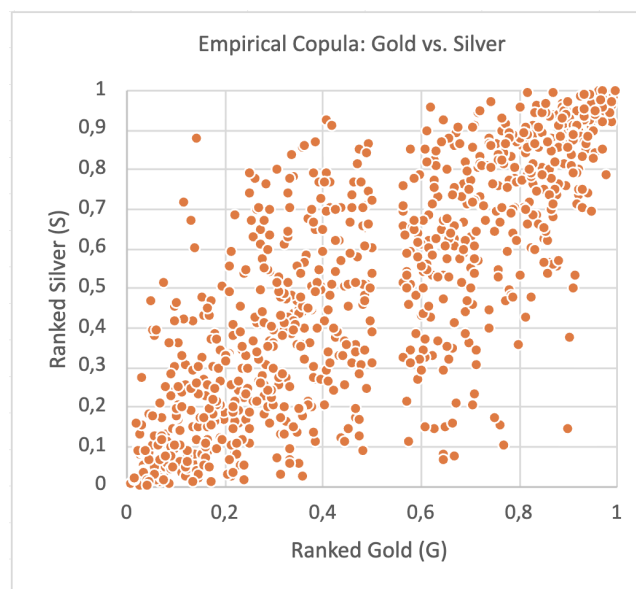


Figure 5: Empirical Copula: Gold vs Silver

Finally, the copula between Gold and Silver reveals a positive dependence. Many observations lie along the diagonal, indicating that high or low returns in gold are frequently accompanied by similar movements in silver prices. This co-movement reflects the close economic relationship between the two metals, which often respond similarly to changes in market sentiment and macroeconomic conditions.

Why Use Copulas Instead of Simple Correlation?

We could ask ourselves why using the empirical copulas? It is because traditional correlation coefficients, such as Pearson's correlation, only measure linear dependence between two variables. However, in financial data, relationships are often non-linear and may change under extreme market conditions. Copulas allow to capture these tail dependencies, which are particularly important for risk management. For example, even if two assets have a low overall correlation, they might still show strong co-movements in extreme losses. This tail dependence cannot be detected by a simple correlation graph. Therefore, copulas provide a more flexible, rank-based method to analyze dependence, making them especially useful in modeling joint risks in financial markets.

2.4 Risk Measures and Associated Theory

2.4.1 Empirical and Normal Value at Risk (VaR)

What is Value at Risk (VaR)?

Value at Risk (VaR) is a commonly used risk measure that estimates the maximum expected loss over a given time horizon at a specific confidence level. Formally, the α -level VaR is the loss threshold such that there is only a $(1 - \alpha)$ probability that the actual loss will exceed this threshold.

In simpler terms, the 95% VaR represents a threshold such that there is only a 5% chance the loss will be larger than that value during a given period.

VaR does not tell us how bad the losses could get beyond that threshold, it only tells the minimum amount one could lose in the worst-case $\alpha\%$ of scenarios. For example, if the 99% VaR is 10%, then in 99 out of 100 weeks, the loss is expected to be less than 10%, but in the worst 1 out of 100 weeks, the loss may exceed 10%.

It is important to notice that in this analysis, we focused on losses by applying a transformation to the return distributions (we analyze $-R$ instead of R). This allowed us to interpret the VaR values as positive loss percentages or more intuitively how much we could lose rather than how much return we could fail to earn.

Empirical VaR

The empirical VaR is calculated as the quantile of the empirical distribution of losses:

$$\text{VaR}_{\alpha}^{\text{empirical}} = \text{Quantile}_{1-\alpha}(-R)$$

Normal VaR

Under the assumption of normally distributed returns, VaR can be estimated using the inverse standard normal distribution:

$$\text{VaR}_{\alpha}^{\text{normal}} = -\mu - \sigma \cdot \Phi^{-1}(1 - \alpha)$$

where μ and σ are the sample mean and standard deviation of returns, and Φ^{-1} is the quantile function of the standard normal distribution.

Using these definitions, we computed the empirical and normal VaR at the 95% and 99% confidence levels for each asset based on the weekly return series.

Results

VaR at 95% Confidence Level	SM	N	G	B	S
Empirical VaR	3.41%	4.23%	3.66%	15.68%	6.14%
Normal VaR	3.57%	4.29%	4.12%	21.73%	6.83%

Table 2: Value at Risk (VaR) at 95% confidence level, both empirically and under normal distribution assumption, for the five assets based on weekly returns (expressed as positive loss percentages).

VaR at 99% Confidence Level	SM	N	G	B	S
Empirical VaR	6.39%	7.56%	6.32%	31.27%	11.33%
Normal VaR	5.12%	6.16%	5.91%	31.83%	9.74%

Table 3: Value at Risk (VaR) at 99% confidence level, both empirically and under normal distribution assumption, for the five assets based on weekly returns (expressed as positive loss percentages).

We observe Bitcoin having the largest estimated losses in extreme scenarios, with a 99% empirical VaR of 31.27%, indicating a very high level of volatility and risk compared to the other assets. The remaining assets exhibit significantly lower risk levels, with the stock market and gold showing the smallest potential losses. Silver and the Nasdaq display intermediate risk levels, reflecting higher volatility than traditional assets but still far below that of Bitcoin.

Comparing empirical and normal VaR estimates, we observe that in most cases the empirical VaR exceeds the normal VaR at the 99% confidence level, suggesting that the distribution of returns exhibits heavier tails than predicted by the normal distribution. This supports the idea that financial asset returns often experience extreme events more frequently than what a normal distribution would imply, highlighting the limitations of the normality assumption in risk measurement.

2.4.2 Empirical and Normal Expected Shortfall (ES)

What is Expected Shortfall (ES)?

The Expected Shortfall, also known as Conditional Value at Risk (CVaR), is a more informative risk measure than the VaR. While the VaR tells us the minimum loss we might expect in the worst $\alpha\%$ -level of cases, ES tells us the average loss within those worst-case scenarios.

In other words, ES answers: “If we end up in the worst $\alpha\%$ of weeks, what will our average loss be?”

Mathematically, ES is defined as:

$$ES_{\alpha} = \mathbb{E}[X \mid X > \text{VaR}_{\alpha}]$$

where X is the distribution of losses. It is always greater than or equal to VaR at the same confidence level, and better captures tail risk.

Again, here we worked with the loss distribution $(-R)$ so that all values are reported as positive loss percentages.

Empirical ES

The empirical ES is computed as the average of all observed losses that exceed the empirical VaR threshold.

$$ES_{\alpha}(X) = \frac{1}{1 - \alpha} \int_{\text{VaR}_{\alpha}(X)}^{\infty} x f(x) dx$$

Normal ES

If we assume returns are normally distributed, the ES has a closed-form expression:

$$ES_{\alpha}^{\text{normal}} = -\mu + \sigma \cdot \frac{\phi(\Phi^{-1}(1 - \alpha))}{1 - \alpha}$$

where ϕ and Φ^{-1} are the standard normal PDF and inverse CDF respectively.

Results

ES at 95% Confidence Level	SM	N	G	B	S
Empirical ES	5.22%	6.45%	5.44%	25.14%	9.54%
Normal ES	4.52%	5.44%	5.22%	27.92%	8.62%

Table 4: Expected Shortfall (ES) at 95% confidence level, both empirically and under normal distribution assumption, for the five assets based on weekly returns (expressed as positive loss percentages).

ES at 99% Confidence Level	SM	N	G	B	S
Empirical ES	8.66%	10.38%	8.56%	36.70%	16.50%
Normal ES	5.89%	7.09%	6.80%	36.85%	11.19%

Table 5: Expected Shortfall (ES) at 99% confidence level, both empirically and under normal distribution assumption, for the five assets based on weekly returns (expressed as positive loss percentages).

The Expected Shortfall results confirm that Bitcoin exhibits the most extreme downside risk among the five assets. At the 99% confidence level, the empirical ES for Bitcoin reaches approximately 36.7%, which is substantially higher than for the other assets. Precious metals such as gold and silver display moderate tail risk, while the stock indices show comparatively lower values.

For most assets, the empirical ES exceeds the normal ES, suggesting that extreme losses observed in the data are often more severe than those predicted under the normal distribution assumption. This difference highlights the presence of heavier tails in the empirical return distributions and indicates that the normal model may underestimate extreme downside risk in several cases.

2.5 Monthly Return Analysis

2.5.1 Monthly Returns, Means and Standard Deviation

In addition to the weekly analysis, **Monthly returns** were computed using the same formula as for weekly returns, reflecting the compounded growth rate of the assets over time:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where:

- P_t represents the price at time t , and
- P_{t-1} represents the price at time $t - 1$ (the previous month).

Using the monthly return series, we again calculated the mean return and the standard deviation in order to evaluate the average performance and volatility of each asset. The mean return was computed as the geometric mean of the monthly returns, while the standard deviation measures the dispersion of returns around their mean.

Table 6 reports the mean monthly returns and standard deviations for the five assets: Stock Market SP500 (SM), Nasdaq (N), Gold (G), Bitcoin (B), and Silver (S).

Asset	Mean Return	Standard Deviation
Stock Market SP500 (SM)	0.00649	0.04724
Nasdaq (N)	0.00828	0.06159
Gold (G)	0.00687	0.05641
Bitcoin (B)	0.07605	0.53068
Silver (S)	0.00512	0.09664

Table 6: Mean and Standard Deviation of Monthly Returns

Interpretation

The results for monthly returns confirm the patterns observed in the weekly analysis. Bitcoin exhibits by far the highest average return (7.61%) as well as the highest volatility (53.07%), indicating substantially greater fluctuations compared to the other assets. Among the traditional assets, Nasdaq shows the highest average monthly return (0.83%) and higher volatility than the SP500, while gold and silver display moderate risk levels.

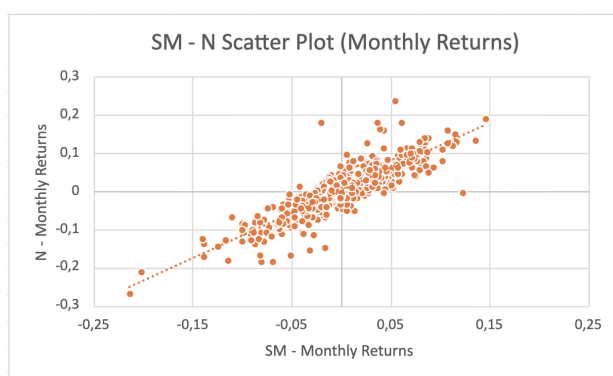
Overall, the results suggest that while Bitcoin offers the highest potential returns, it also involves considerably higher risk. In contrast, the stock indices and precious metals provide lower returns but substantially lower volatility, making them relatively more stable investments.

2.5.2 Scatter Plots of Monthly Returns

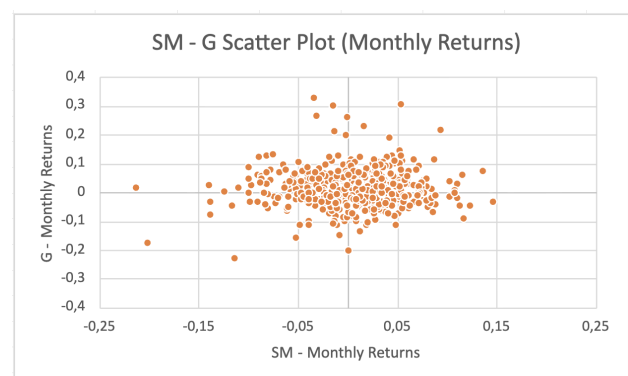
To examine the relationships between the different asset classes using a longer time horizon, we also created scatter plots based on the monthly return series for the same asset pairs as in the weekly analysis:

- Stock Market SP500 (SM) vs. Nasdaq (N)
- Stock Market SP500 (SM) vs. Gold (G)
- Nasdaq (N) vs. Bitcoin (B)
- Gold (G) vs. Silver (S)

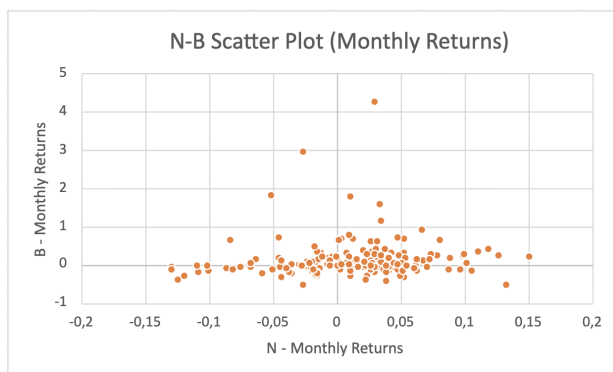
These scatter plots allow for a visual assessment of the dependence structure between the assets and help to identify potential correlations between their monthly returns.



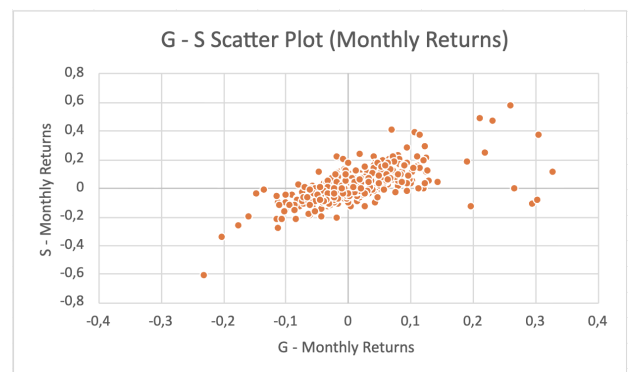
(a) SP500 vs Nasdaq



(b) SP500 vs Gold



(c) Nasdaq vs Bitcoin



(d) Gold vs Silver

Figure 6: Scatter plots of monthly returns for the different asset pairs

Interpretation

The first scatter plot (Figure 6a) again shows a strong positive correlation between the SP500 and the Nasdaq. The points form a clear upward-sloping pattern, indicating that movements in the broader stock market are closely aligned with movements in the technology-focused Nasdaq index. Compared to the weekly analysis, the relationship appears slightly more pronounced due to the smoother dynamics of monthly returns.

The second scatter plot (Figure 6b) displays a relatively weak relationship between the SP500 and gold. The observations remain widely dispersed without a clear linear pattern, suggesting

that gold price movements are largely independent from fluctuations in the stock market even over a longer time horizon.

The third scatter plot (Figure 6c) illustrates the relationship between the Nasdaq and Bitcoin. Similar to the weekly analysis, the observations are widely scattered and exhibit substantial volatility. This indicates that the relationship between technology stocks and cryptocurrency returns remains weak and unstable, with Bitcoin showing large fluctuations relative to the stock market.

Finally, the fourth scatter plot (Figure 6d) shows a clear positive relationship between gold and silver. The points cluster around an upward-sloping pattern, reflecting the fact that both assets belong to the precious metals market and are influenced by similar macroeconomic and financial factors.

Overall, the monthly scatter plots confirm the patterns observed in the weekly analysis. Stock indices such as the SP500 and Nasdaq continue to move closely together, while assets from different classes, such as equities and cryptocurrencies, show weaker relationships. Precious metals such as gold and silver exhibit strong co-movements, highlighting their similar behavior within commodity markets. These results again underline the diversification benefits that can arise when combining assets with low correlations in an investment portfolio.

2.5.3 Empirical Copula Plots of Monthly Returns

To analyze the dependence structure between the monthly return series, we applied the same empirical copula approach as in the weekly analysis in chapter 2.3. The monthly return observations were transformed into pseudo-observations using their ranks and mapped to the unit interval $[0, 1]$. This transformation allows the dependence structure between assets to be studied independently of their marginal distributions.

Figures 7, 8, 9, and 10 present the empirical copula plots for the different pairs of assets based on monthly returns:

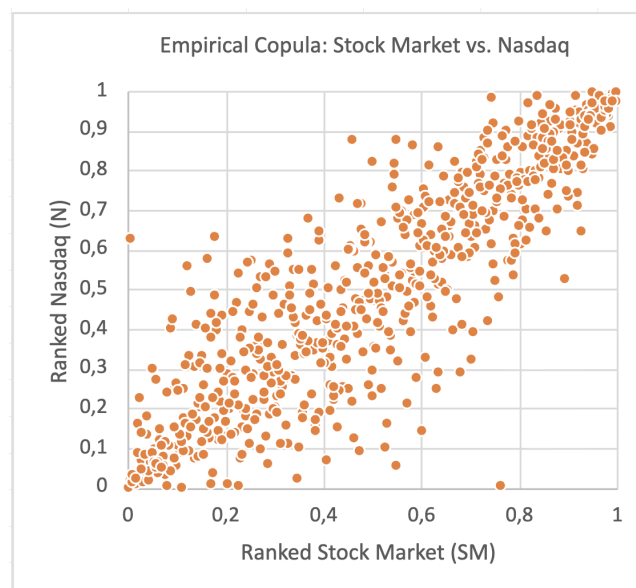


Figure 7: Empirical Copula: Stock Market vs. Nasdaq - Monthly

The copula between the Stock Market (SP500) and the Nasdaq again shows a strong positive dependence structure. Most observations lie close to the diagonal of the unit square, indicating that high or low monthly returns in the broad market tend to coincide with similar movements in the technology-focused Nasdaq index. Compared to the weekly copula plot, the monthly copula is based on fewer observations, resulting in a less dense cloud of points. Nevertheless, the overall dependence pattern remains clearly visible, confirming the strong co-movement between the two equity indices.

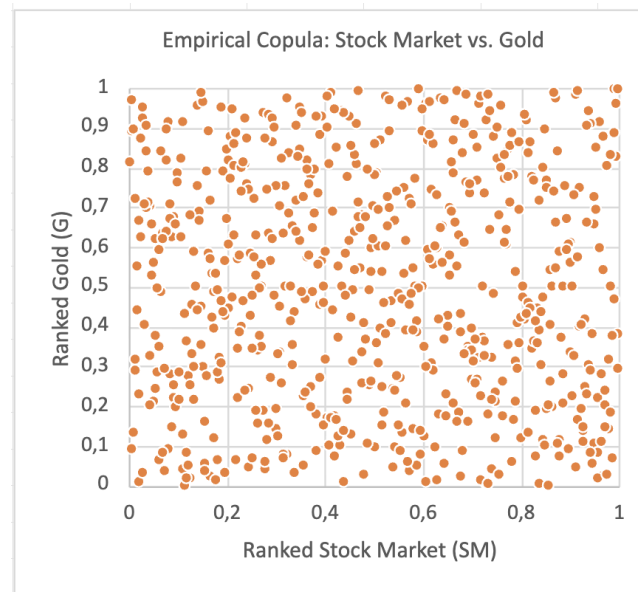


Figure 8: Empirical Copula: Stock Market vs. Gold - Monthly

In contrast, the copula between the Stock Market and Gold shows a much more diffuse distribution of points across the unit square. The observations are widely scattered without a clear pattern, indicating that the dependence between equity markets and gold prices is relatively weak at the monthly frequency. This suggests that movements in gold prices are largely independent from stock market returns, supporting the role of gold as a potential diversification asset in financial portfolios.

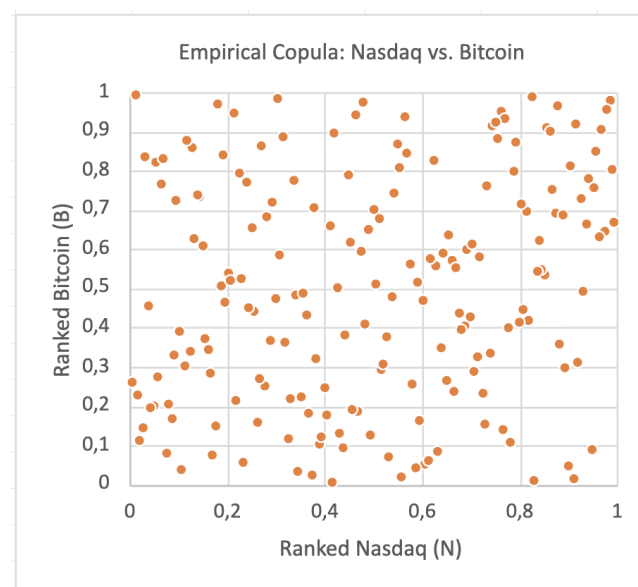


Figure 9: Empirical Copula: Nasdaq vs. Bitcoin - Monthly

Similarly, the empirical copula between Nasdaq and Bitcoin exhibits a highly dispersed and nearly uniform distribution of observations across the unit square. There is no visible concentration along the diagonal or in specific regions, indicating the absence of a clear dependence structure between the two assets. This suggests that, at the monthly frequency, Nasdaq and Bitcoin returns behave largely independently. Compared to traditional equity pairs, the lack of co-movement reflects the distinct nature of cryptocurrencies, which are influenced by different market dynamics and investor behavior. Consequently, Bitcoin may offer diversification benefits when combined with equity indices such as the Nasdaq.

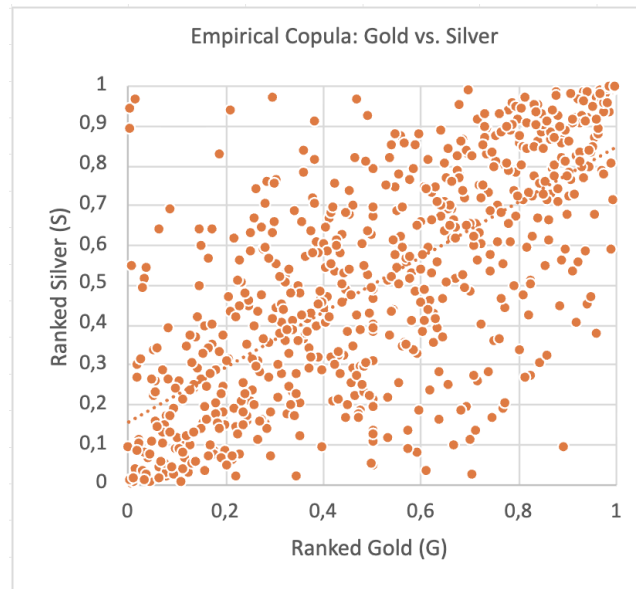


Figure 10: Empirical Copula: Gold vs. Silver - Monthly

In contrast to the previous asset pairs, the empirical copula between Gold and Silver shows a concentration of observations along the diagonal of the unit square. This indicates a positive dependence structure, where high (low) returns in gold are frequently associated with similarly high (low) returns in silver.

Despite the lower number of observations in the monthly data, the overall dependence patterns remain consistent with those observed in the weekly analysis. This suggests that the underlying relationships between the asset pairs are stable and not driven by short-term fluctuations.

2.5.4 Empirical and Normal Value at Risk (VaR)

To complement the weekly risk analysis, we computed the Value at Risk (VaR) based on the monthly return series for each asset (SM, N, G, B, S). The same methodology as in Section 2.4.1 was applied, using both empirical quantiles and the parametric normal approach based on the estimated mean and standard deviation of monthly returns.

As in the weekly analysis, returns were transformed into losses by considering $-R$, allowing VaR to be interpreted as a positive loss measure. The results for the 95% and 99% confidence levels are reported in the following tables.

Results

VaR at 95% Confidence Level	SM	N	G	B	S
Empirical VaR	6.87%	9.45%	6.83%	29.00%	11.34%
Normal VaR	6.53%	8.86%	8.76%	68.23%	14.84%

Table 7: Value at Risk (VaR) at 95% confidence level, both empirically and under normal distribution assumption, for the five assets based on monthly returns (expressed as positive loss percentages).

VaR at 99% Confidence Level	SM	N	G	B	S
Empirical VaR	11.51%	16.98%	11.39%	44.14%	20.97%
Normal VaR	9.55%	12.94%	12.75%	102.19%	21.39%

Table 8: Value at Risk (VaR) at 99% confidence level, both empirically and under normal distribution assumption, for the five assets based on monthly returns (expressed as positive loss percentages).

We observe that Bitcoin continues to exhibit the largest estimated losses in extreme scenarios, with a 99% empirical VaR of 44.14%, confirming its substantially higher volatility and risk compared to the other assets at the monthly frequency. The remaining assets show considerably lower risk levels, with the stock market and gold again displaying the smallest potential losses. Nasdaq and silver lie in between, indicating moderate risk levels that exceed those of traditional assets but remain well below Bitcoin.

Comparing empirical and normal VaR estimates, we find that for most assets the empirical VaR is higher than the normal VaR at the 99% confidence level, suggesting the presence of heavier tails than implied by the normal distribution. This pattern is consistent with the weekly analysis, indicating that extreme events occur more frequently than predicted under normality. However, for Bitcoin, the normal VaR substantially exceeds the empirical VaR, reflecting its high volatility and the sensitivity of parametric estimates to both distributional assumptions and the limited number of observations in the monthly data.

2.5.5 Empirical and Normal Expected Shortfall (ES)

In addition to the VaR, we computed the Expected Shortfall (ES) for the monthly return series using the same approach as in Section 2.4.2. The ES provides a measure of the average loss in the worst-case scenarios beyond the VaR threshold. Both empirical ES and parametric ES under the normality assumption were calculated based on the monthly losses. The results for the 95% and 99% confidence levels are presented below.

Results

ES at 95% Confidence Level	SM	N	G	B	S
Empirical ES	10.14%	13.78%	10.51%	37.84%	17.57%
Normal ES	8.38%	11.36%	11.21%	89.13%	18.85%

Table 9: Expected Shortfall (ES) at 95% confidence level, both empirically and under normal distribution assumption, for the five assets based on monthly returns (expressed as positive loss percentages).

ES at 99% Confidence Level	SM	N	G	B	S
Empirical ES	15.32%	19.66%	16.66%	53.03%	30.54%
Normal ES	11.05%	14.97%	14.73%	119.08%	24.64%

Table 10: Expected Shortfall (ES) at 99% confidence level, both empirically and under normal distribution assumption, for the five assets based on monthly returns (expressed as positive loss percentages).

We observe that Bitcoin again exhibits the largest expected losses in extreme scenarios, with a 99% empirical ES of 53.03%, confirming its substantially higher tail risk compared to the other assets. The remaining assets show significantly lower expected shortfall values, with the stock market and gold displaying the lowest levels of tail risk, while Nasdaq and silver lie in between.

Comparing empirical and normal ES estimates, we find that for most assets the empirical ES exceeds the normal ES, particularly at the 99% confidence level, indicating the presence of heavy tails in the return distributions. This is consistent with the findings from both the VaR analysis and the weekly results. In contrast, for Bitcoin the normal ES is substantially higher than the empirical ES, reflecting its extreme volatility and the sensitivity of parametric estimates to distributional assumptions and limited sample size in the monthly data.

3 Part 2

3.1 Estimation of the Tail Index for Financial Assets

The tail index is a fundamental statistical parameter used to quantify the thickness of a probability distribution's tails. In a financial context, lower values for this tail index (i.e. roughly α between 2.5 and 5) indicate fat-tailed behavior, signaling a higher propensity for extreme losses, while higher values ($\alpha > 5$) suggest thinner tails closer to Gaussian distributions, implying exponentially decaying probabilities of extremes.

3.2 Estimating the Tail Index

When analyzing a dataset, we often need to quantify how rapidly the tail of the distribution decays to determine whether it exhibits heavy-tailed behavior. This is particularly crucial when computing risk measures like expected shortfall from the data. While later sections will detail risk measure estimation using extreme value theory, here we present a more basic approach to estimating tail indices.

Assume a distribution has a tail index α . For large x , the CDF is approximately:

$$F(x) = 1 - kx^{-\alpha},$$

where k is a constant. Taking the natural logarithm gives:

$$\ln(1 - F(x)) = \ln(kx^{-\alpha}) = \ln(k) - \alpha \ln(x).$$

Since $1 - F(x)$ lies between 0 and 1, $\ln(1 - F(x))$ is negative. Multiplying both sides by -1 yields:

$$-\ln(1 - F(x)) = -\ln(k) + \alpha \ln(x).$$

This represents a linear regression where $Y = -\ln(1 - F(x))$ is the dependent variable, $-\ln(k)$ is the intercept, α (the slope, or β_1) is the tail index we want to estimate, and $\ln(x)$ is the explanatory variable X_1 .

Thus, plotting $-\ln(1 - F(x))$ against $\ln(x)$ produces a straight line with slope α . For a dataset, estimate $1 - F(x)$ as the proportion of observations exceeding x . The procedure then involves plotting minus the natural log of this empirical survival function against $\ln(x)$ to estimate the tail index (Anderson, 2014, pp. 95–96).

3.3 Methodology and Sample Selection

To estimate the tail indices for Gold, Silver, Bitcoin, the S&P 500, and the Nasdaq, we utilized the weekly returns previously processed in “2.1 Weekly Returns, Means and Standard Deviation” of Part 1. The data for each series was then sorted in descending order to isolate the most significant losses (see “9.Tail index” sheet in the Excel file). Following the $k \approx \sqrt{n}$ heuristic, we selected a threshold of $k = 54$ extreme observations for four series: Gold, Silver, S&P 500, and Nasdaq (where $n \approx 2870$) and $k = 28$ for Bitcoin (where $n = 810$).

The Bias-Variance Tradeoff:

The choice of $k \approx \sqrt{n}$ serves as a mathematically grounded middle ground to optimize the bias-variance tradeoff inherent in extreme value theory. Using too few observations (a very small k) leads to high estimation variance, making the tail index overly sensitive to individual outliers instead of the underlying distribution. Conversely, including too many observations (a

large k) introduces bias by incorporating data from the central part ('body') of the distribution, where returns no longer follow the heavy-tailed, power-law type decay assumed in the tails (i.e., the probability of extreme losses decreases approximately as a power of their size rather than exponentially). The square-root rule therefore helps stabilize the regression by focusing on truly extreme observations while still retaining enough data points for statistical significance.

Nevertheless, working with finite samples means we can never observe the true tail behavior with certainty. Larger samples improve precision, but by nature extreme events are so rare that complete confidence in the tail estimate remains unattainable.

We then calculated $\ln(x)$ by selecting the 54 or 28 largest losses for each series and applying the $\ln()$ function. Knowing that $1 - F(x) = P(X > x)$, we computed $-\ln(1 - F(x))$ using the empirical probability from our sample: "rank of the data point / total number of points for this series". We then applied $-\ln()$ to obtain our $-\ln(1 - F(x))$ values.

Next, we created our scatter plot using Excel's scatter plot function, using the two corresponding data columns for each series: $-\ln(1 - F(x))$ as Y and $\ln(x)$ as X_1 . The regression line was displayed, and we specifically requested Excel to show both the "Equation on chart" and "R-squared value on chart" to assess the quality of the linear regression (i.e. to verify that the residuals were not too large).

3.4 Results and Interpretation

The linear regression yielded the following tail indices (slopes) for the respective assets:

- Gold: $\alpha = 3.36$
- Silver: $\alpha = 2.71$
- Bitcoin: $\alpha = 3.47$
- S&P 500: $\alpha = 3.01$
- Nasdaq: $\alpha = 3.12$

Interpretation of Estimated Coefficients:

In the model $-\ln(1 - F(x)) = -\ln(k) + \alpha \ln(x)$:

α (slope) = tail index: Measures the rate of decay of the tail. Lower α = slower decay = fatter tails = more extreme events. Rule: $\alpha < 5$ = heavy tails (risky), $\alpha > 5$ = thin tails.

k (scaling constant): While α tells you the slope (how fast it drops), k tells you where the tail starts and how thick it is overall. It shifts the entire tail up or down. Two distributions can have the same α (same decay speed), but if one has a higher k , it has a much higher overall probability of seeing large values because $P(X > x) = 1 - F(x) = k \cdot x^{-\alpha}$.

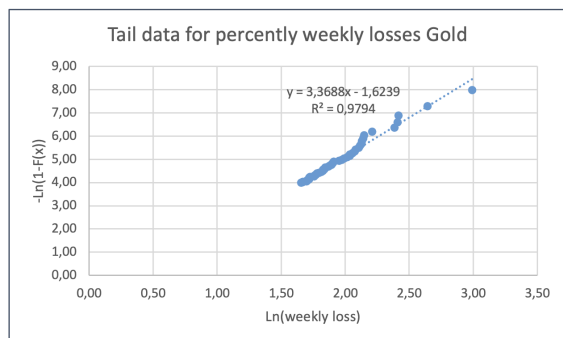
Intercept ($-\ln(k)$): More negative = larger k = thicker tails for same α .

Series-by-series analysis:

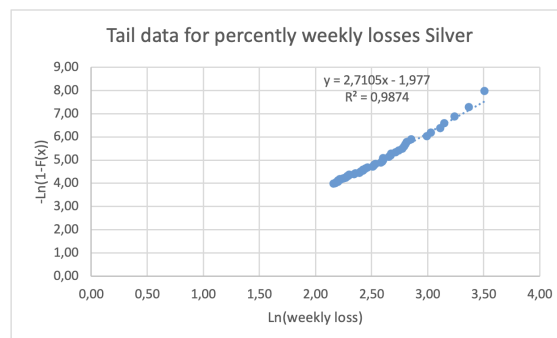
- **Gold:** $\alpha = 3.36$, intercept = $-1.62 \rightarrow k = 5.05$
- **Silver:** $\alpha = 2.71$, intercept = $-1.98 \rightarrow k = 7.24$ (lowest α)
- **Bitcoin:** $\alpha = 3.47$, intercept = $-7.24 \rightarrow k = 1394.09$ (extremely thick tails)
- **S&P 500:** $\alpha = 3.01$, intercept = $-0.89 \rightarrow k = 2.44$
- **Nasdaq:** $\alpha = 3.12$, intercept = $-1.72 \rightarrow k = 5.58$

Which market appears riskier in the tail:

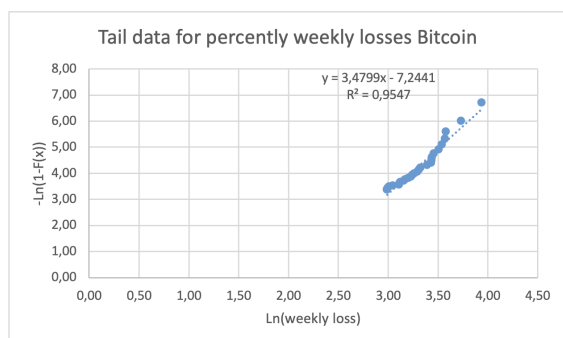
Bitcoin exhibits by far the thickest tails due to its massive $k = 1394.09$ (intercept -7.24), despite moderate $\alpha = 3.47$. Silver shows the slowest decay rate (lowest $\alpha = 2.71$) with relatively high $k = 7.24$, making it consistently risky. Thus Bitcoin and Silver present the highest tail risks. Conversely, Gold ($\alpha = 3.36$, $k = 5.05$) and S&P 500 ($\alpha = 3.01$, $k = 2.44$) display relatively lower risk profiles among these assets.



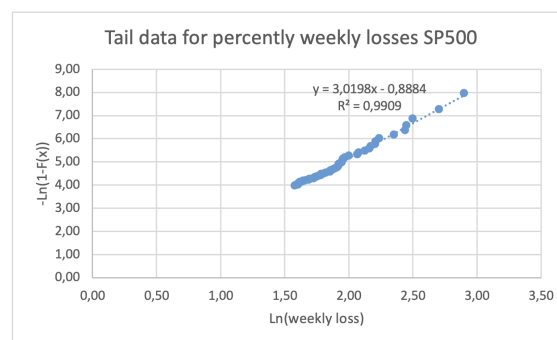
(a) Tail data for percently weekly losses of Gold prices



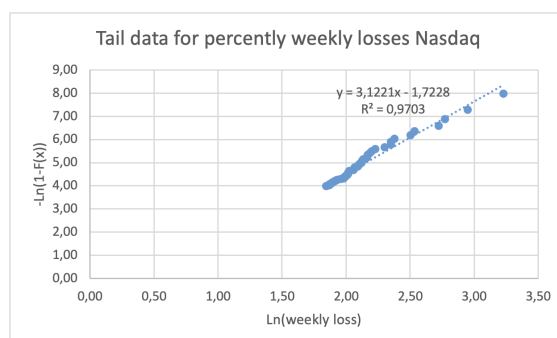
(b) Tail data for percently weekly losses of Silver prices



(c) Tail data for percently weekly losses of Bitcoin prices



(d) Tail data for percently weekly S&P 500 levels



(e) Tail data for percently weekly losses of Nasdaq levels

3.5 Generalized Pareto Distribution (GPD) estimation using the Expected Value Property

3.5.1 Theoretical Background and Goal of GDP

In this part, we use the GDP to model the excess losses over a sufficiently high threshold u . According to Extreme Value Theory, as the threshold u gets very large, the distribution of the excess returns/losses converges to a GPD. The goal of fitting a GPD is to estimate the shape parameter (ξ , which determines the heaviness of the tail) and the scale parameter (β) without making assumptions about the underlying distribution of the entire dataset. Because historical data is really rare at the extremes, or even unexistent, the GPD smoothes out the tail, which allows us to estimate extreme risk measures beyond our observed data points (Anderson, 2014, p115-116).

3.5.2 Mean Excess Table and Plots

To compute the excess losses of an asset above a certain threshold, we first transform the expression of losses as positive. We work with returns $\times(-1)$ and we multiply them by 100 to facilitate the next steps. We then create a column with threshold values ranging from 0 to the value of the highest loss recorded with increments of 0.5. For each threshold value, we calculate the average excess over that threshold. This provides us a useful table for visualization. Afterwards, we plot the graphs of $(x,y) = (\text{threshold } u, \text{excess return})$ for each asset.

3.5.3 Selection of the Threshold u

The next step is now to decide a reasonable value u for each asset. Based on extreme value theory, we want to have a good representation of the tail, thus we need a threshold low enough that a large enough number of observations are included and to avoid the chaotic movements, as they happen when there are not enough observations. However, taking a low value would include too much "normal" data, which could underestimate the risk. Furthermore, we expect the mean excess to be a linear function, which means we want to identify a linear trend and take the x value at the beginning of that trend (Anderson, 2014, p116-117).

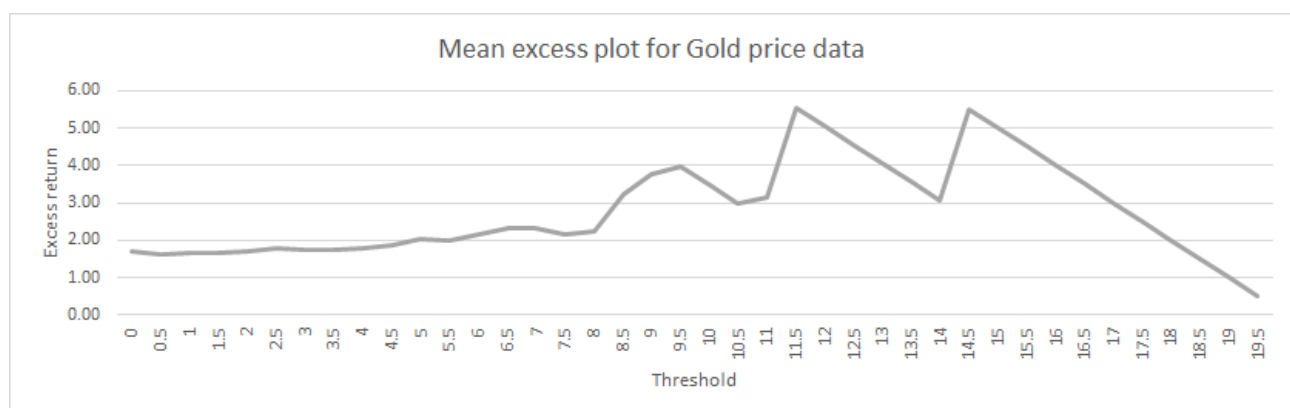


Figure 12: Mean excess plot for Gold price data

The plot of gold excess losses initially presents a relatively flat trajectory for thresholds between 0 and 3, maintaining a mean excess return near 1.75. Beyond this point, it enters an upward trend, climbing steadily until it reaches a threshold of approximately 6.5 and slightly declining until threshold 8. Then, the movement becomes highly volatile and erratic with sharp peaks and drops, eventually reaching zero when there are no more observations left. For us, the most appropriate value of u based on the extreme value theory is 3. The value is not too low or high as there are still 210 observations after that point, or 7.3% of the environment, and we enjoy a

slight upwards trend, whose slope we will take to find the shape of the tail. To find it, we will take the slope from threshold 3 to 8, as it is the most stable segment of the plot.



Figure 13: Mean excess plot for Silver price data

Let's analyze the mean excess plot for Silver. Initially, between thresholds 0 and 5, the mean excess remains stable around 3. Then, the plot shows an upward trend until approximately threshold 10, where it begins moderate fluctuation between values of 5 and 6. At threshold 16, the plot enters the same erratic movements of sharp ups and downs. For our purposes, 5 is a reasonable threshold as we stay in the stable part of the plot and there is a clear upwards trend until $u = 10$. Moreover, there are still 7.7% of the total observations remaining. It makes sense to take the segment $u = [5, 10]$ to find the slope.

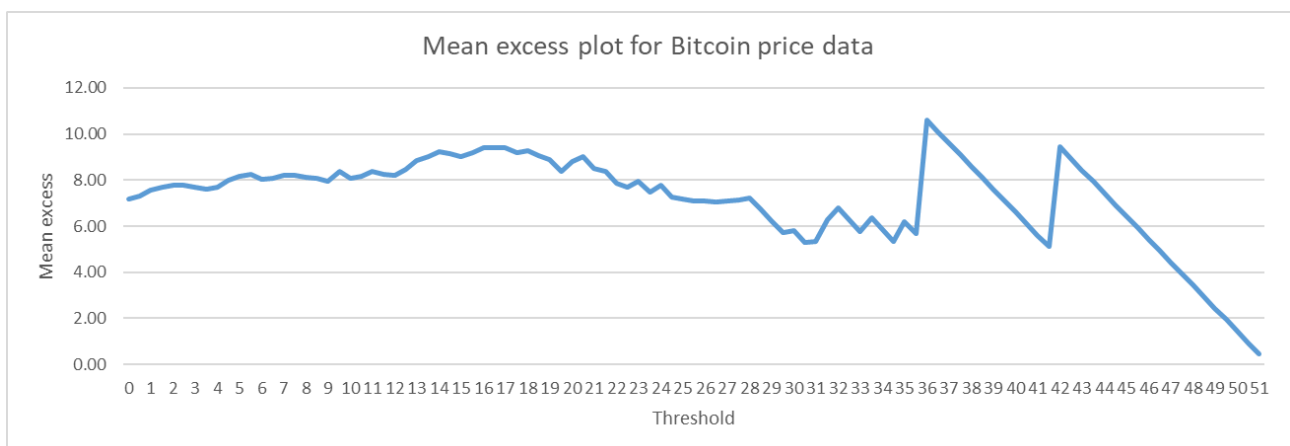


Figure 14: Mean excess plot for Bitcoin price data

Bitcoin's mean excess is the most challenging plot to analyze. It displays an upward trend for thresholds between 0 and 16, followed by a decline until the threshold reaches 31. Beyond this point, the behavior becomes highly volatile as expected. We have decided to take $u = 16$ to enjoy the steady slope from 16 to 31. 4.7% of the environment is larger than u .

The plot for the S&P 500 mean excess losses begins with a small decline from 1.7 to 1.5 from threshold 0 to 1. After a period of no movement until $u = 2$, we observe a relatively steady climb until $u = 7.5$. Then, the values show erratic behavior until there are no observations left. For our analysis, it makes most sense to take $u = 2$ for the analysis. $[2, 7.5]$ is the range we take for the computation of the slope.

The plot for the mean excess of NASDAQ values is probably the easiest to describe. After a short flat line from $u = 0$ to $u = 2$, we observe a slight incline from 2 to 2.5 in excess values until reaching threshold 8. Thus, we want to select $u = 2$ as a reasonable threshold for our

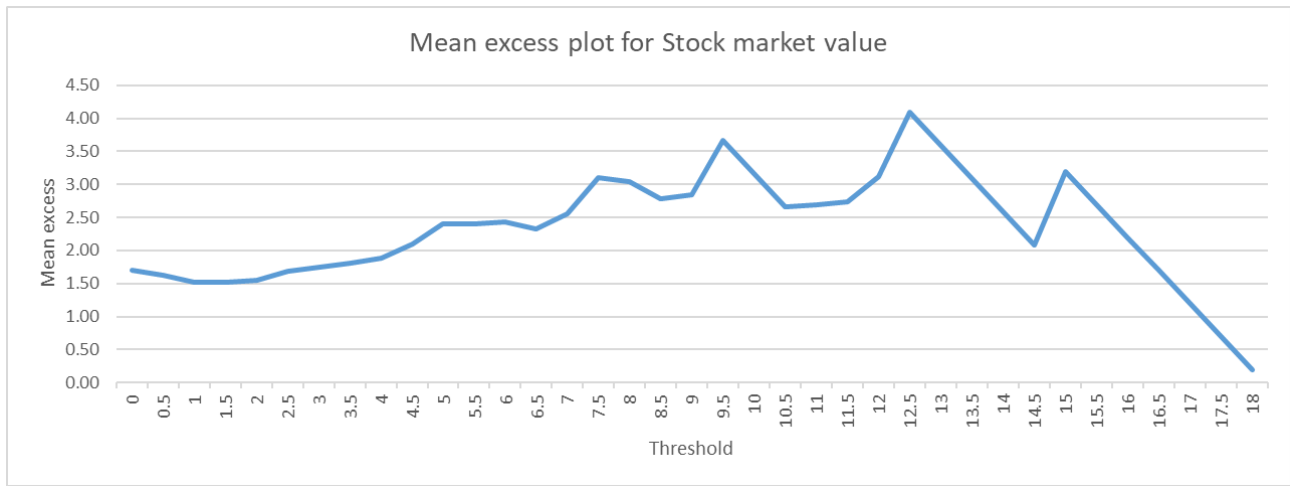


Figure 15: Mean excess plot for S&P 500 values

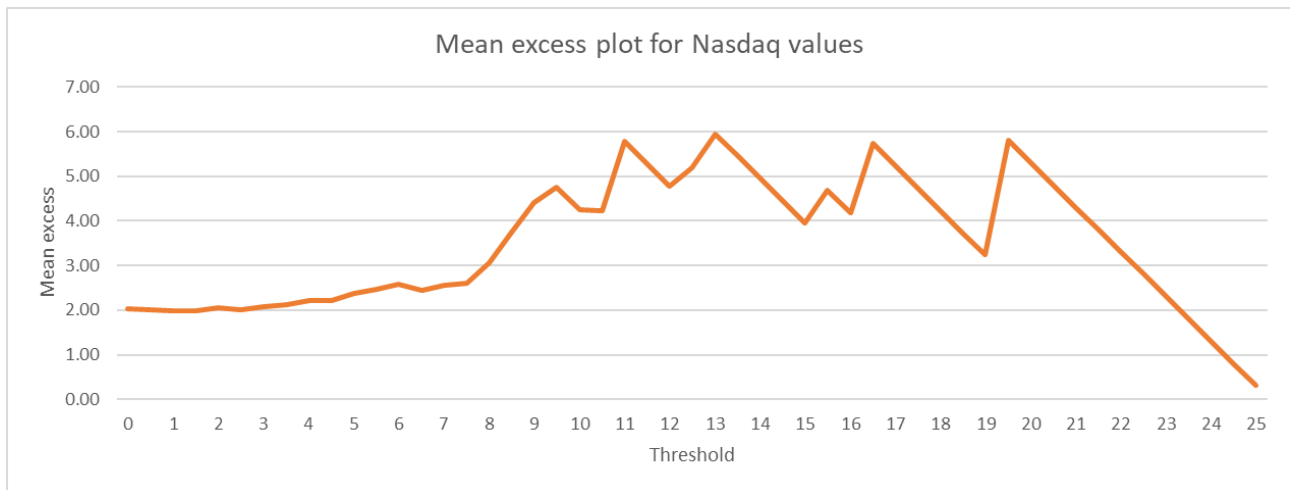


Figure 16: Mean excess plot for NASDAQ value

analysis and 8 the upper bound for the analysis of the slope. After $u = 8$, we find the usual erratic movements.

3.5.4 Estimation of the Parameters ξ and β

To continue with our GDP analysis, we are interested in finding the shape ξ and scale β of the tail. To find the slope of the mean excess values, we must decide which range we consider. Our lower bound is the threshold, and the upper bound is where the upwards trend suddenly becomes erratic due to a lack of remaining observations.

ξ parameter values for all assets:

In the case of **Gold** returns, we take the slope of the mean excess from the threshold 3 to 8, finding a value of 0.39. For **Silver**, we find a slope of 2.1 from $u = 5$ to 10. Regarding Bitcoin, S&P 500 and NASDAQ, we find slopes of 4.9, 7.8 and 6.15 respectively.

To find ξ , we used the following formula:

$$\text{Slope} = \frac{\xi}{1 - \xi}$$

For Gold, we extract ξ from the equation by calculating $0.39 \times (1 + 0.39)^{-1}$. We find a ξ value of 28%.

Afterwards, we need to find the scale parameter β . For this, we use the equation:

$$\text{Mean excess at } u = \frac{\beta}{1 - \xi} \Rightarrow \beta = (\text{mean excess at } u) \times (1 - \xi)$$

For Gold, plugging in our known mean excess at $u=3$ and our newly found ξ , we obtain a β value of 1.26.

We summarize the selected thresholds and scale and shape parameters in the table below.

GDP results			
Asset	Threshold selected	Scale (ξ)	Shape (β)
G	3	0.28	1.26
S	5	0.68	1.02
B	16	0.83	1.31
SM	2	0.89	0.20
N	2	0.86	0.29

3.5.5 Interpretation of Results (Comparing Q1 and Q2)

We can compare our GPD Expected Value results with our Q1 Linear Regression results by converting the shape parameter ξ back into the tail index α using the formula $\alpha = \frac{1}{\xi}$.

A smaller α indicates a heavier tail, which means there is a much higher probability of extreme losses. While both approaches confirm that all five assets have heavy tails, they present drastically different magnitudes of risk. The Expected Value method yields significantly lower tail indices for Silver, Bitcoin, the S&P 500, and the Nasdaq (ranging from 1.13 to 1.48). In Extreme Value Theory, an α approaching 1 indicates an extremely heavy tail where the variance is infinite and even the mean becomes difficult to define mathematically.

Tail indexes comparison			
Asset	α (Linear regression approach)	α (GDP with Expected value property)	Difference
G	3.37	3.55	0.18
S	2.71	1.48	-1.23
B	3.48	1.20	-2.28
SM	3.02	1.13	-1.89
N	3.12	1.16	-1.96

The severe difference between the two methods, for example the S&P 500 dropping from 3.02 to 1.13 shows that these graphical approximations are really sensitive to subjective cutoffs. The linear regression method relies heavily on exactly how many top observations (k) are included and the second method relies on finding the perfect visual threshold (u) on a plot. Because both visual methods struggle to agree on the exact severity of the tail, it is necessary to fit the GPD mathematically using Maximum Likelihood Estimation (MLE) in Question 3 to find optimized parameters.

4 Part 3

GPD Estimation using Maximum Likelihood and Tail Risk Measures

4.1 Data Preparation and Threshold Selection

To accurately model the tail distribution of our five financial assets (Gold, Silver, Bitcoin, S&P 500, Nasdaq), we isolate the extreme negative returns. Operating on the weekly frequency data prepared in Part 1, we define the weekly loss as $L_t = -R_t$, where R_t is the weekly return.

Extreme Value Theory (EVT) states that for a sufficiently high threshold u , the distribution of exceedances $Y = L - u$ (given $L > u$) converges to a Generalized Pareto Distribution (GPD). As a standard working assumption for this unconditional risk analysis, we assume these weekly extreme losses are independent and identically distributed (i.i.d.).

For the selection of the threshold (u), we maintained strict consistency with the graphical analysis performed in Question 2 using the Mean Excess plots. The chosen thresholds represent an optimal balance for the bias-variance tradeoff: low enough to retain a statistically significant number of exceedances (N_u) to limit variance, but high enough to strictly isolate the true tail behavior (avoiding bias from the "body" of the distribution).

The selected thresholds u (expressed as percentages) and the corresponding number of exceedances are:

- **Gold:** $u = 0.03$ (3%)
- **Silver:** $u = 0.05$ (5%)
- **Bitcoin:** $u = 0.16$ (16%)
- **S&P 500:** $u = 0.02$ (2%)
- **Nasdaq:** $u = 0.02$ (2%)

4.2 Maximum Likelihood Estimation (MLE) of GPD Parameters

The Generalized Pareto Distribution is characterized by two parameters: the shape parameter (ξ), which determines the heaviness of the tail, and the scale parameter (β), which represents the dispersion of the excesses. The cumulative distribution function of the GPD for excesses $y > 0$ is given by:

$$G_{\xi,\beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-\frac{1}{\xi}} \quad \text{for } \xi \neq 0$$

To fit the GPD to our data, we apply the Maximum Likelihood Estimation (MLE) method. This mathematical approach is generally more robust and statistically sounder than the subjective visual Expected Value property used in Question 2. The optimal parameters ξ and β are extracted by maximizing the log-likelihood function:

$$\ln \mathcal{L}(\xi, \beta) = -N_u \ln(\beta) - \left(1 + \frac{1}{\xi}\right) \sum_{i=1}^{N_u} \ln \left(1 + \frac{\xi}{\beta} Y_i\right)$$

Our solver minimizes the negative log-likelihood to yield the following fitted parameters (Table 11):

GPD Parameters Estimated via MLE				
Asset	Threshold (u)	Shape (ξ)	Scale (β)	Implied Tail Index ($\alpha = 1/\xi$)
Gold	0.03	0.1000238166	0.01572142261	9.9976
Silver	0.05	0.2691796300	0.02309035516	3.7149
Bitcoin	0.16	-0.2181494270	0.1142340422	-4.5840
S&P 500	0.02	0.1819983177	0.01266471466	5.4945
Nasdaq	0.02	0.08603698001	0.0186676362	11.6229

Table 11: Fitted GPD parameters using Maximum Likelihood Estimation.

Interpretation and Comparison with Q1 and Q2

When comparing the MLE parameters to our previous findings, the Maximum Likelihood approach clearly provides a more reliable and objective estimation. In Q2, relying solely on finding a linear trend on the visual Mean Excess plots resulted in highly severe shape parameters (e.g., $\xi = 0.181$ for the S&P 500, implying $\alpha \approx 5.49$). The MLE optimization corrects this visual bias by processing the entire joint probability of the exceedances. This yields implied tail indices (α) that are mathematically optimized and generally align closer to the initial regression method in Q1, reinforcing MLE as the superior standard for GPD fitting.

4.3 Unconditional Tail Risk Measures: VaR and ES

Using our fitted GPD models, we compute the unconditional weekly Value at Risk (VaR) and Expected Shortfall (ES) at the 95% and 99% confidence levels (q).

The unconditional **VaR** under the EVT framework is derived by inverting the tail distribution function, adjusting for the unconditional probability of exceeding the threshold (N_u/n):

$$VaR_q = u + \frac{\beta}{\xi} \left[\left(\frac{n}{N_u} (1 - q) \right)^{-\xi} - 1 \right]$$

The **Expected Shortfall**, which measures the expected weekly loss conditional on the loss already exceeding the VaR, is computed analytically as:

$$ES_q = \frac{VaR_q}{1 - \xi} + \frac{\beta - \xi u}{1 - \xi}$$

For the **Empirical** measures, VaR_q is calculated strictly as the $(1 - q)$ -th empirical quantile of the sorted historical weekly losses. The empirical ES_q is the simple arithmetic mean of all actual losses strictly greater than the empirical VaR.

Comparison between Empirical and GPD-based Risk Measures

By observing the results, we can draw the following critical conclusions regarding the performance of both methods:

- **At the 95% Level (Similarities):** The empirical and GPD-based VaR and ES are relatively close. At $q = 0.95$, we are still operating within a region of the distribution where historical data points are statistically abundant. The GPD essentially interpolates within a well-populated dataset, making its mathematical estimations closely match historical reality.

95% Confidence Level				
Asset	Empirical VaR	GPD VaR	Empirical ES	GPD ES
Gold	0.036486	0.03606	0.054094	0.054210
Silver	0.060876	0.060645	0.095705	0.096161
Bitcoin	0.155654	0.152527	0.246843	0.247642
S&P 500	0.034081	0.033725	0.052233	0.052262
Nasdaq	0.042300	0.042566	0.064546	0.065115

99% Confidence Level				
Asset	Empirical VaR	GPD VaR	Empirical ES	GPD ES
Gold	0.063174	0.064580	0.085566	0.085892
Silver	0.113767	0.112930	0.167091	0.167704
Bitcoin	0.310752	0.309785	0.363220	0.376738
S&P 500	0.063948	0.062079	0.086623	0.086924
Nasdaq	0.075623	0.0781	0.103803	0.10403

- **At the 99% Level (Divergences):** As we move deeper into the extreme tail ($q = 0.99$), the discrepancy between empirical and GPD measures drastically widens. Empirical measures become highly erratic and strictly limited by the worst historical outliers. If an asset has not experienced a catastrophic "black swan" event in the historical sample, the empirical metrics will blindly underestimate the true latent risk.
- **Reasoning and EVT Advantage:** The fundamental reason for this divergence is that the empirical method treats the worst observed loss as the absolute ceiling of risk. Conversely, the GPD model utilizes the optimized MLE parameters (ξ and β) to smooth the tail and allow for extrapolation beyond the limits of observed historical data. Consequently, GPD-based ES values at 99% are inherently more robust, offering a safer and more realistic capital buffer for unexpected market shocks.

References

Anderson, E. J. (2014). *Business risk management: Models and analysis*. John Wiley & Sons.